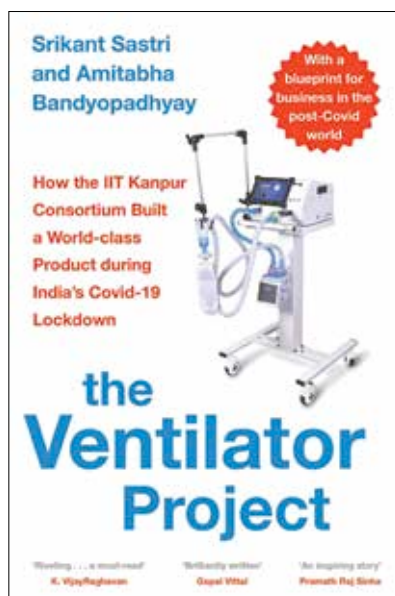


queries and we also received many offers for collaboration. The movement grew to become a committed network of volunteers from all walks of life, determined to do their bit for the country.” The team started working on the project in earnest, adding a new feature every three days.

One of the problems was that there was a lack of clarity on the technical requirements and the features required for the ICU ventilators. These requirements were just not there, and throughout the development process, new features were being added to the product. Note that the Union Health Ministry officially prescribed the minimum requirements for COVID-19 ventilators in August, a month after Noccarc started the first commercial shipments of its world class made in India ventilator. By that time, other Indian companies had already developed, manufactured and delivered ventilators. Throughout the time, Noccarc was rapidly iterating, developing viable versions of their ventilator. We asked Nikhil and Harshit how the features on their ventilator was picked, “Initial features were selected based on the requirements and specifications laid out by UK and Canadian governments. The prototypes, NOCCARC V110 & NOCCARC V210, were developed in line with these specifications. Government of India also released a set of specifications at a later stage through an HLL tender. NOCCARC V310 was developed keeping in consideration the specifications laid down in this tender.”

Towards the later stages of the development, when bringing the product to the market, the engineers who had previously worked on making robots for cleaning solar panels without the use of water, were for the first time, going through the hoops required to get a medical device certified and tested in India. According to Nikhil and Harshit, things could have been easier, “There are gaps in the regulations related to manufacturing and selling of medical devices in India. There is a strong need for clear and well-laid down regulations and guidelines. At the moment, these are unclear and hazy. There are many



The Ventilator Project has all the details of the development of the ICU ventilator by the IIT Kanpur incubator

tests and certifications mentioned in the regulatory guidelines. However, the regulatory body or organisation also needs to declare a list of test labs and certifying bodies within India. This will help beneficiaries easily navigate and conform to regulatory compliances. Similarly, with the advancement of technology, test labs and certification bodies must be made available in the country to make the process smooth and hassle-free for innovators. This will also help the country become self-sufficient in its product development journey. Along with strong and clear regulatory guidelines, a mechanism to check the genuineness and authenticity of certifications must also be laid down. This will help ensure that the market is not at a risk from the inflow of inauthentic products.”

We asked if the ventilator was used on COVID-19 patients during the development, “No. The Noccarc team was absolutely certain that it would not use the ventilator on any patient till it was thoroughly tested on test lungs under supervision of doctors and validated by them. It was only the final tested, validated and certified ventilator that was first used on COVID patients. Dr Prachee Sathe from Ruby Clinic helped us at crucial steps on the way to product development.

She provided feedback on the ventilator and the product’s functionalities. The Noccarc team was also aided at Ruby Clinic for endurance testing and clinical testing of the ventilator. The same was approved to be fit for use in the ICU on 26th June 2020.”

The Ventilator Project is stuffed to the brim with handy tips for entrepreneurs, and a whole framework for accelerating the development of world class products. Licensing of the designs for production by other manufacturers is one of the ways a startup can increase capacity. To manufacture Noccarc, the team had to encounter a number of shady encounters with competing manufacturers who wanted to steal or look at the designs, and had trouble finding an established company to manufacture the ventilator according to the specifications. During this time, the team looked at SMEs and even PSUs to license their designs to.

Among the many ideas presented in the course of the book for boosting the manufacturing sector in India is one that suggests incubators, accelerators or leadership programs set up speed dating events between startups and SMEs. The authors say, “start-ups at the product development stage need to focus on open innovation. Hence, it becomes crucial for them to collaborate with companies that can fill in the missing pieces of the product development puzzle. However, the key lies in finding appropriate partners - that is where incubators and accelerators step in. Bridging gaps between start-ups and SMEs makes SME’s user base accessible for the start-up and provides SMEs with an outlet for their developed manufacturing and supply chain networks. Start-ups have the right talent and investor backing to create a pipeline of innovations. On the other hand, SMEs have the innate skills to manage the complex business environment in India, especially related to manufacturing (where issues of land, labour and legislation can be quite challenging).” After looking around for suitable partners to manufacture the device, the Noccarc team settled on Avi Healthcare,